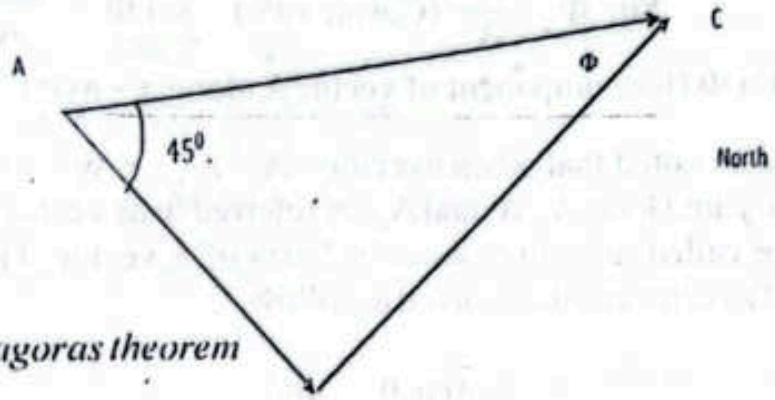
Solution

The first displacement is represented by a line AB drawn 45° east of north, 3 unit long and with an arrow on it to show the direction of travel. From the head of line we draw the second displacement BC as a line 4 units long, 40°west of north i.e. at right angles to AB. AC represents the resultant sum of the two vectors AB and BC both in magnitude and direction. Measurement shows that the resultant displacement is 5 km in a direction of 8° west of north.



Using Pythagoras theorem

$$R^2 = 3^2 + 4^2 = 25$$

$$R = \sqrt{25} \quad 5 \text{ unit}$$

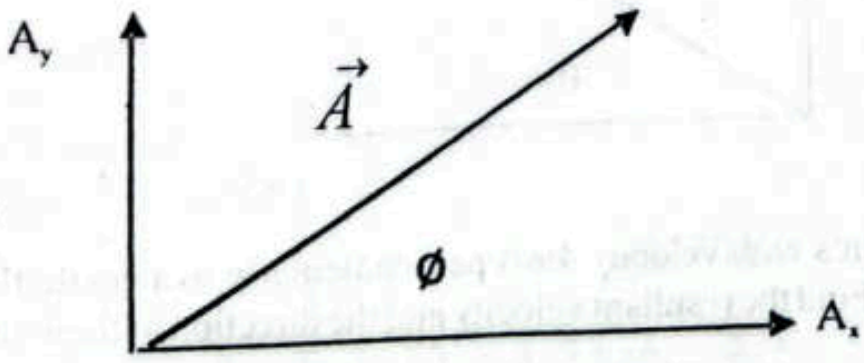
Also, using sine rule to find the angle(direction)

$$\frac{\text{Sin}(\Phi + 45^{\circ})}{4} = \frac{\sin 90^{\circ}}{5}$$

$$\Phi = 8^{\circ}$$

2.6 Component of a Vector

A component of a vector $\vec{A}$ *is the projection of the vector along the axis.* The proper description of a vector can be formed through its components. Consider figure below, a projection



of a vector $\vec{A}$ along the x-axis with the x-component represented by A_x along the y-axis with the y-component represents by A_y. We can find the x and y component by converting the polar coordinates to rectangular coordinates using cosine and sine identities:

$$\text{Cos } \Phi = \frac{Ax}{A} \text{ (Cosine rule)}$$

$A_x = A \cos \Phi$ (for component of vector $\vec{A}$ along x - axis)

$$\text{Sin } \Phi = \frac{Ay}{A} \text{ (Cosine rule)} \quad \sin \Phi = \frac{A_y}{A}$$

$A_y = A \sin \Phi$ (for component of vector $\vec{A}$ along x - axis)

Also, it should be noted that when avectors $A = A_x i + Ay\, i + Azk$ with the coefficient of i, j amd k i.e. A_x, A_y and A_z are referred to as vector components , i, j and k are called unit directional or basis of a vector. The x- and y-components of a vector can be resolved as follow:

$$A_x = A\cos\theta \quad \text{and}$$

$$A_y = A\sin\theta$$

Also, the magnitude and the direction of a vector $\vec{A}$ relate through the equations below:

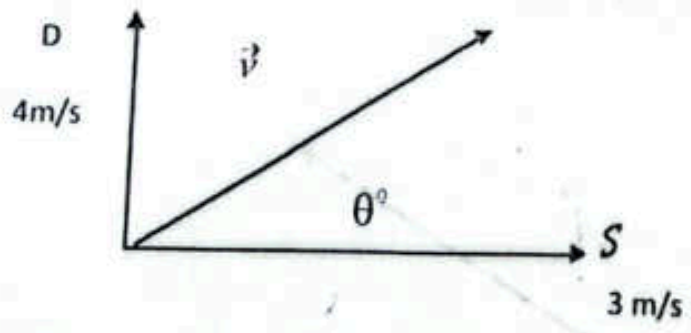
$$A^2 = A_X^2 + A_Y^2 \text{ (Pythagoras theorem)}$$

$$A = \sqrt{A_x^2 + A_y^2} \text{ (For magnitude of vector A)}$$

$$\tan\theta \quad \frac{A_y}{A_x} \text{ (tangent rule)}$$

$$\phi \quad \tan^{-1}\frac{A}{A_x} \text{ (for direction of vector } \vec{A}\text{)}$$

Example 2.5



A diver D, swims with velocity 4m/s perpendicularly to a gentle flowing of sea S, at 3m/s. Find the resultant velocity and the direction of the diver

$$V^2 = V_X^2 + V_Y^2$$

$$V^2 = 3^2 + 4^2 \text{ (Pythagorean theorem)}$$

$$A = \sqrt{3^2 + 4^2} = 5\text{m/s}$$

$$\tan\phi = \frac{A_y}{A_x} \text{ (tangent rule)}$$

$$\tan\phi = \frac{4}{3}$$

$$\phi \ \text{Tan}^{-1}\left(\frac{4}{3}\right) = 53^{\circ},$$

hence, 5m/s, to S - Direction

2.7 Magnitude of a Vector

Given the vectors $\vec{V} = V_x i + V_Y j + V_z$ ks, the magnitude of vector $\vec{V}$ can be obtained by applying Pythagoras theorem, so that

$$|\vec{V}| = V = \sqrt{(V_x)^2 + (V_y)^2 + (V_z)^2}$$

Example 2.6:

Given that $\vec{V} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{U} = \hat{i}, \ 2\hat{j} + 3\hat{k}$, find the magnitude of $\vec{V}$ and $\vec{U}$

Solution For a vector $\vec{V} = 3i + 2j + k$

$$|\vec{V}| = V = \sqrt{(3)^2 + (2)^2 + (1)^2}$$

$$|\vec{V}| = V = \sqrt{9 + 4 + 1}., \text{ then, } |\vec{V}| = \sqrt{14}$$

For vector $\vec{U} = \hat{i}, \ 2\hat{j} + 4\hat{k}$

$$|\vec{U}| = U = \sqrt{(1)^2 + (,\ 2)^2 + (4)^2}$$

$$|\hat{U}| = U = \sqrt{1 + 4 + 16} = \sqrt{21}$$

2.8 Addition and Subtraction of Vectors

Given the vectors $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$ addition and subtraction of the two vectors can be achieved by adding or subtracting the components of . and . so that

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}.$$

$$\vec{A}, \ \vec{B} = (A_x, \ B_x)\hat{i} + (A_y, \ B_y)\hat{j} + (A_z, \ B_z)\hat{k}.$$

Example 2.7

Given that vector $\vec{A} = 3\hat{i} + 2\hat{j}, \hat{k}$ and $\vec{B} = 2\hat{i} + 2\hat{j} + 4\hat{k}$. Find i. $(\vec{A} + \vec{B})$ ii. $(\vec{A}, \vec{B})$

Solution

i $(A \text{ and } B) = (3 + 2)(2+2)\,j + (-1 + 4)\,k$

Then, $(\vec{A} + \vec{B}) = 5\hat{i} + 4\hat{j} + 3\hat{k}$

Ii $(\vec{A}, \vec{B}) = (3, 2)\hat{i} + (2, 2)\hat{j} + (, 1, 4)\hat{k}.$

$(\vec{A}, \vec{B}) = \hat{i}, 5\hat{k}.$

2.8 Unit Vector

Given the vector $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ the unit vector of vector A is given by

$$\hat{n} = \frac{\vec{A}}{|\vec{A}|}$$

Example 2.8

Given the $\vec{V} = 3\hat{i} + 2\hat{j} + \hat{k}$ find the unit vector of vector $\vec{V}$.

Solution

Using: $\hat{n} = \frac{\vec{V}}{|\vec{V}|}$

To find the magnitude of $V = 3\hat{i} + 2\hat{j} + \hat{k}$, we have:

$$|\vec{V}| = \sqrt{(3)^2 + (2)^2 + (1)^2}$$

$|\vec{V}| = \sqrt{14}$ hence

$$= \frac{\vec{V}}{|\vec{V}|} = \frac{3\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{14}}$$

The unit vector of vector V, becomes $\hat{V} = \frac{3}{\sqrt{14}}\hat{i} + \frac{2}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}$

2.9 Multiplication of a Vector

There are two main categories of vector multiplication known as:

i. Scalar (Dot) product of a Vector

ii. Cross (Vector) product of a Vector.

The scalar multiplication of vectors produced scalar product while the vector multiplication of vectors formed vector product. However, the scalar-vector multiplication of vectors yields vector product.

Scalar (Dot) Product of Vectors

Given two vectors:

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k} \text{ and}$$

$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k},$$

A scalar or dot product of the two vectors can be obtained by multiplying the components of i, j and k, as follow:

$$\vec{A}/\vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) \cdot (B_x\hat{i} + B_y\hat{j} + B_z\hat{k})$$

$$\vec{A}/\vec{B} = (A_x \times B_x) + (A_y \times B_y) + (A_z \times B_z),$$

Note that $\hat{i}/\hat{i} = \hat{j}/\hat{j} = \hat{k}/\hat{k} = 1$ and $\hat{i}/\hat{j} = \hat{j}/\hat{k} = \hat{i}/\hat{k} = 0$

Similarly, the angle between two vectors $\vec{A}$ and $\vec{B}$ is given by "

$$\vec{A} / \vec{B} = |\vec{A}|\ |\vec{B}| \cos\theta$$

$$\cos\theta = \frac{\vec{A}.\vec{B}}{|\vec{A}|\,|\vec{B}|}$$

$$\theta = \cos^{-1}\left(\frac{\vec{A}.\vec{B}}{|\vec{A}|\ |\vec{B}|}\right)$$

Example 2.9

Given the two forces $\vec{F}_1 = \hat{i} + 2\hat{j} \quad 4\hat{k}$ and $\vec{F}_2 = 3\hat{i}\ \hat{j} + 5\hat{k}$ find the scalar Product of vector $\vec{F}_1$ and $\vec{F}_2$ hence calculate the angle between $\vec{F}_1$ and $\vec{F}_2$

$$\vec{F}_1\ \vec{F}_2 = (\hat{i} + 2\hat{j} - 4\hat{k}) \cdot (3\hat{i} - \hat{j} + 5\hat{k})$$

$$\vec{F}_1\ \vec{F}_2 = (1 \times 3) + (2 \times (-1)) + ((-4) \times 5)$$

$$\vec{F}_1.\vec{F}_2 = 3 - 2 - 20$$

$$\vec{F}_1\ \vec{F}_2 = -21$$

Magnitude of vector $\vec{F}_1$ and $\vec{F}_2$ are:

$$\vec{F}_1 = \sqrt{(1)^2+(2)^2+(-4)^2}$$

$$\vec{F}_1 = \sqrt{21}$$

$$\vec{F}_2 = \sqrt{(3)^2+(-1)^2+(5)^2}$$

$$\vec{F}_2 = \sqrt{35}$$

To find angle between $\vec{F}_1$ and $\vec{F}_2$ are:

$$\vec{F}_1 . \vec{F}_2 = |\vec{F}_1 \vec{F}_2| \cos\theta$$

$$\theta = \cos^{-1}\left(\frac{\vec{F}_1 \ \vec{F}_2}{|\vec{F}_1| \ |\vec{F}_2|}\right)$$

$$\theta = \cos^{-1}\left(\frac{-21}{\sqrt{21}\sqrt{35}}\right)$$

$$\theta = 141^\circ$$

Cross (Vector) Product of Vectors

Given two vectors $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$ the cross product of the two vectors can be obtained by arranging the components of $\hat{i}, \hat{j}$ and $\hat{k}$ in a 3 by 3 matrices and obtaining the determinant of the matrix as vector or cross product

$$\vec{A} \times \vec{B} = \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{pmatrix}$$

$$\vec{A} \times \vec{B} = \hat{i}\begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} - \hat{j}\begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix} + \hat{k}\begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix}$$

$$\vec{B} = \hat{i}(A_yB_z - A_zB_y) - \hat{j}(A_xB_z - A_zB_x) + \hat{k}(A_xB_y - A_yB_x)$$

Example 2.10.

Given the two forces $\vec{F}_1 = \hat{i} + 2\hat{j} - 4\hat{k}$ and $\vec{F}_2 = 3\hat{i} - \hat{j} + 5\hat{k}$ and . find the cross product of vector $\vec{F}_1$ and $\vec{F}_2$.

Solution

Cross product of $\vec{F}_1$ and $\vec{F}_2$

$$\vec{F}_1 \times \vec{F}_2 = \begin{vmatrix} + & - & + \\ i & j & k \\ 1 & 2 & -4 \\ 3 & -1 & 5 \end{vmatrix}$$

$$\vec{F}_1 \times \vec{F}_2 = i\begin{vmatrix} 2 & -4 \\ -1 & 5 \end{vmatrix} - j\begin{vmatrix} 1 & -4 \\ 3 & 5 \end{vmatrix} + k\begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix}$$

$$\vec{F}_1 \times \vec{F}_2 = i(10 - 4) - j(5 + 12) + k(-1 - 6)$$

$$\vec{F}_1 \times \vec{F}_2 = 6\hat{i} - 17\hat{j}, 7\hat{k}$$

Exercise 2

1 Given that $\vec{m} = 2\hat{i} - 3\hat{j}$ - $\vec{n} = \hat{i} - 2\hat{k}$ and$\vec{o} = 3\hat{i} + 2\hat{j} + 4\hat{k}$. Show that:

i m n = n m (*Ans*:2)

ii m + n + o = n + o + m (*Ans*: 61 - j + 2k)

iii Find the angle between m and n (*Ans*: 88.3°)

2 Given two vectors $\vec{A} = 6\hat{i} + 3\hat{j} - \hat{k}$ and$\vec{B} = 4\hat{i}$, $5\hat{j} + 8\hat{k}$ Find the magnitude of the vector 2A - 3B [*Ans*: 43.54]

3. Given that $\vec{X} = 6\hat{i} - 2\hat{j} - \hat{k}$ and $\vec{Y} = 2\hat{i} + 4\hat{j} - 3\hat{k}$ find the magnitude of X and Y (**Ans:**$\sqrt{42}$ **and**$\sqrt{29}$)

4. Given that vector $\vec{A} = \hat{i} + 2\hat{j} + 5\hat{k}$ and$\vec{B} = 4\hat{i} - 2\hat{j} + \hat{k}$, find i. $(\vec{A} + \vec{B})$ ii. $(\vec{A} - \vec{B})$

(*Ans*: 5i + 6k and -3i + 4j + 4k)

5. Given the two forces A = 6i - 3j-4k and B = 2i + j + 5k. , find the scalar product of vector F_1 and F_2 hence calculate the angle between F_1 and F_2 [Ans: -12 and 106.3°]

6. Given the two forces F_1 = 6i - 3j-4k and F_2 = 2i + j + 5k, find the vector or cross product of vector F_1 and F_2 [Ans: 11i - 38j + 36k]

7. Vectors K and L are given as $\vec{K} = 3\hat{i} + 4\hat{j} - 2\hat{k}$ and$\vec{L} = 2\hat{i} - 6\hat{j} + 3\hat{k}$ Find the angle between K and L [129.5°]

8. The resultant of two vectors **X** and **Y** is $Z_r = 2.2i + 3.4j$, if vector . X = 1.5 - 2.0j Find the Magnitude of Y and the angle it makes with the positive x- axis. [*And*: 5.45 Unit, 82.6°].

9. Given two vectors X = 2i + 3j + 4k and Y = i - 2j + 3k. What is the magnitude of each vector?[***Ans:* 5.4 and 3.7 unit]**

10. Given a vector **A** of magnitude 4 units, lying in the x-y plane and makes an angle 30° of with the x-axis. Calculate the components along the x, y and z-axes.[Ans: 2 3, 2 , 0]

11. If x= 2i+3j , y = 3i+5j and z = 4i+6j. Find $x \times y \times z$ **[Ans: 6i+4j]**

CHAPTER THREE

PARTICLES IN MOTION

3.1 KINEMATICS

Mechanics is a branch of physics concerned with the motion of bodies under the action of forces; it's usually divided into two parts: kinematics and dynamics. *Kinematics deals with the mathematical description of the motion of objects without consideration of what causes the motion only the displacement, velocity and acceleration are considered.* Dynamics, on the other hand, study the causes of the motion. We will begin the study of mechanics with kinematics of particles.

A **particle** is a body whose dimensions, except mass are negligible.

Motion means a continuous change in the position of a body relative to a reference point, as measured by a particular observer in a particular frame of reference. In other words, motion is any physical movement or change in position or place.

3.1.1 Types of Motion

1. **Rectilinear motion**: This is a type of motion, which takes place along a straight line. It is also known as translational and linear motion; an example is a man running a race on a straight track.

2. **Circular motion**: This is the motion of a body travelling in a circle. For example:
   - A ball tied to a spring and whirled around
   - Motion of electron about the nucleus
   - Movement of the moon around the earth.

3. **Random motion**: This is the motion whose direction and speed keeps changing continuously, it is also a haphazard or irregular non-directed type of motion. Example of such motion are; the movement of gas molecules, a flying butterfly, etc.

4. **Rotational motion**: This is the type of motion in which an object rotates or spins on its own axis. For example; motion of a turntable, rotation of the earth about its axis causing day and night.

5. **Oscillatory motion**: This is a to and fro motion of a particle in a steady manner. Familiar examples of oscillatory motion are; the motion of a simple pendulum, the motion of a vertically loaded stretched

spring. alternating current power.

3.1.2 Some Basic Definitions

1. **Distance:** This is the interval or Separation between two given points in a straight line. S.I unit of distance is meters (m).

2. **Displacement:** This is defined as the change in position of an object. It is a vector quantity and has magnitude and direction Displacement can also be defined as the distance travel in a direction

3. **Speed:** This can be defined as the rate of change of distance with time. Mathematically

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{time taken}}$$

S.I unit: m/s

4 **Uniform speed:** A body is said to have uniform speed if it covers equal distances in equal time intervals with zero acceleration.

5. **Average speed:** this is the ratio of the total distance travelled to the total time taken.

$$\text{Average Speech} = \frac{\text{total distance travelled}}{\text{total time taken}} = \frac{x}{t}$$

6 **Velocity:** This is the rate of change of displacement with respect to time. S.I unit: m/s

$$\text{Velocity} = \frac{\text{Displacement}}{\text{time}} = \frac{x}{t}$$

7. **Uniform velocity:** This is when the rate of change of displacement with respect to time is constant.

8. **Average velocity:** This is the ratio of total displacement to the total time taken.

$$\text{AverageVelocity} = \frac{\text{Tota displacement}}{\text{time taken}} = \frac{x_f - x_i}{t_f - t_i} = \frac{\Delta x}{\Delta t}$$

Where x_f & x_i are the final and initial displacements respectively t_f & t_i are the time taken

9. **Acceleration:** This is rate of change of velocity with respect with time.

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{time taken}} = \frac{v_f - v_i}{t_f - t_i} = \frac{\Delta v}{\Delta t}$$

Where v_f & v_i are the final and initial velocity respectively v_f & v_i are the time taken

10. **Uniform acceleration:** This is when the rate of change of velocity with respect with time is constant

11. **Retardation:** This is defined as the decrease in velocity per unit time. That is retardation occurs when the velocity of a moving body decreases. Retardation is regarded as the negative of acceleration or deceleration

3.2 EQUATIONS OF MOTION

Consider a body undergoing a constant acceleration motion in which the rate of change of velocity is constant throughout the motion.in which the body would start with an initial velocity u and moving with a constant acceleration a for a time t at the end of which it attains a velocity v after Covering a distance s. By definition,

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{time taken}}$$

$$a = \frac{v-u}{t} \quad \text{Rearranging the equation}$$

$$v = u+at \ldots\ldots\ldots\

From equation 1, we have

$$t = \frac{v - u}{a} \quad \text{.....(2.5)}$$

Substituting equation 2.5 into equation 1.5

$$S = \left(\frac{v + u}{2}\right)\left(\frac{v - u}{a}\right)$$

$$v^2 = u^2 + 2as \text{.....(3)}$$

Therefore, the four (4) equations of motion are:

1. $v - u + at$
2. $s = ut + \frac{1}{2} at^2$
3. $v^2 = u^2 + 2as$
4. $S = \left(\frac{v+u}{2}\right) t$

Solved examples

1. A body initially travelling at 10m/s is given a retardation of $2m/s^2$ for 4s. Find its final velocity.

Solution

$u = 10m/s$, $a = 2m/s^2$, t 4s, $v = ?$

$V = u + at = 10 - 2(4) = 2m/s.$

1. A race car starting from rest accelerates at a constant rate of $5.00m/s^2$ What is the velocity of the car after it has travelled 1.00×10^2 ft? (b) How much time has elapsed? (c) Calculate the average velocity. (d) Suppose the driver slams on the brake, stopping the car in 4.00s. Find (i) the acceleration (ii) the distance the car travels while braking, assuming the acceleration is constant. (iii) the average velocity while braking.

Solution

To concert the distance 1.00×10^2 ft to m, is important to recall that 1m = 3.281ft

$$1.00 \times 10^2 ft = 1.00 \times 10^2 ft \times \frac{1m}{3.281ft} = 30.5m$$

(a) To calculate the velocity of the car after it has travel 1.00×10^2 ft, equation (3) can be applied

$v^2 = u^2 + 2as$.....(3)

$$V = \sqrt{(0)^2 + 2\left(5.00 \frac{m}{s^2}\right)(30.50m)} = 17.5m/s.$$

(a) To calculate the time elapsed apply equation (1)

$$v = u + at$$

$$t = \frac{v - u}{a} = \frac{17.5m/s}{5.00/s^2} = 3.50s$$

(c) Average velocity = $\frac{x_f - x_i}{t_f - t_i}$

$$\frac{30.5 - 0}{3.5 - 0} = 8.71m/s$$

(a) (i) When the brake is apply, the final velocity gradually turn zero acceleration can then be then be calculated using

$$v = u + at$$

$$O - 17.5m/s + (a \times 4)$$

$$a = -4.38m/s^2$$

(ii) equation (3) or (4) can be used to determine the distance traveled when braking with constant acceleration. Using equation (3)

a, $v^2 = u^2 + 2as$

$$s = \frac{v^2 - u^2}{2a} = \frac{-(17.5)^2}{2 \times (-4.38)} = 34.96m$$

Apply equation (4) to confirm the answer

(iii) average speed during the deceleration

$$\frac{\text{distance traved during deceleration}}{\text{Time taken}}$$

$$\frac{34.96m}{4.00m} = 8.74m/sm$$

2. A car decelerates with $10m/s^2$ from 50m/s to 20m/s.Calculate the distance travelled by the car.

Solution

$a = -10m/s^2$, $v = 20m/s$ and $u = 50m/s$, $s = ?$

$v^2 = u^2 + 2as$

$20^2 = 50^2 - (2 \times (-10) \times s)$

$20^2 = 50^2 - 20s$

$s = 105m$

3. A vehicle is moving at 5m/s. how long will It take to stop distance of 20m?

Solution

$v = 0, u = 5\text{m/s}, s = 20\text{m}$

$0^2 = 5^2 + (2 \quad (-a) \quad 20)$

$a = 25/40 = 0.625\text{m/s}^2$

From $v = u + (-a)\, t$

$$t = \frac{v - u}{a} = \frac{0 - 5}{-0.625} = 8s$$

4. An electric train moving at 15 km/h accelerates to a speed of 30 km/h in 20s. Find the distance travelled in meters during the period of acceleration.

Solution

1km = 1000m. 1 hour = 3600s

$$u = \frac{15\text{km}}{\text{h}} = \frac{15000\text{m}}{3600\text{s}} = 4.17\text{m/s}$$

$$v = \frac{30\text{km}}{\text{h}} = \frac{30000}{3600} = 8.33\ \text{m/s}$$

$$u = \frac{v - u}{t} = \frac{8.33 - 4.17}{20} = 0.208\text{m/s}^2$$

$$s = ut + \frac{1}{2}at^2$$

$$s = 4.17 \times 20 + \frac{1}{2} \times 0.208 \times 20^2$$

$$s = 83.4 + 41.6 \quad 125\text{m}$$

5 A train travelling at 20m/s undergoes a uniform retardation of 2m/s^2, when the brakes are applied. Calculate the time taken to come to rest and the distance traveled from the place where the brakes were applied.

Solution

$u = 20\text{m/s}, a = -2\text{m/s}^2$ and $v = 0$

a). $v = u + at$

$0 = 20 - 2t$

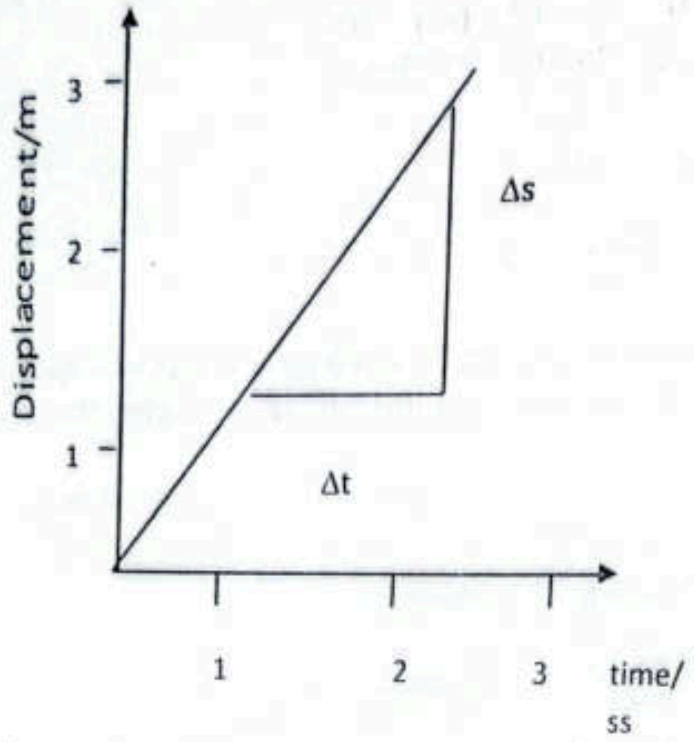
$t = 10s$

b). the distance, $s = ut + \frac{1}{2}at^2$

$$s = 20 \times 10 - \frac{1}{2} \times 2 \quad 102 = 100\text{m}$$

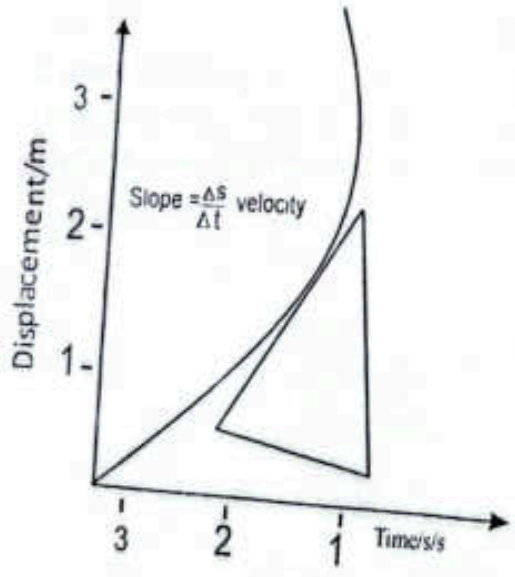
3.2.1 GRAPHS OF MOTION

1. Displacement time graph

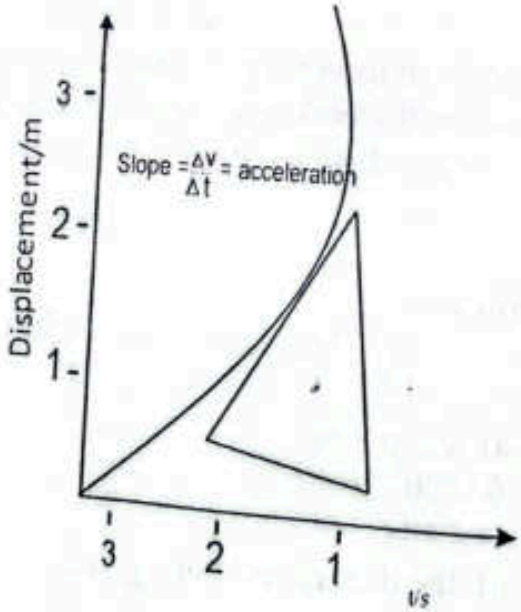


For a body moving with non-uniform velocity, the graph is a curve and the velocity at any time is called **instantaneous velocity**.

Instantaneous velocity at any instant is defined as the gradient of the tangent to the displacement-time curve at that instant.

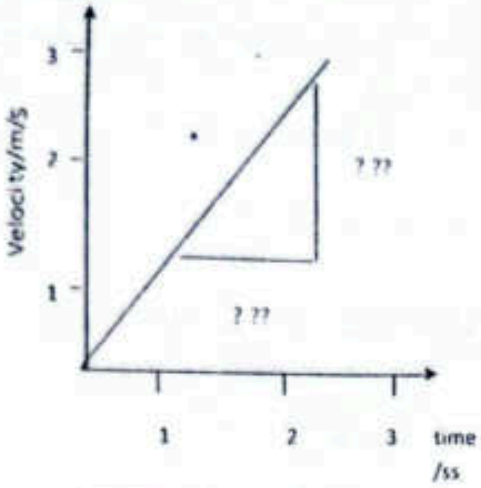


Non - uniform Velocity

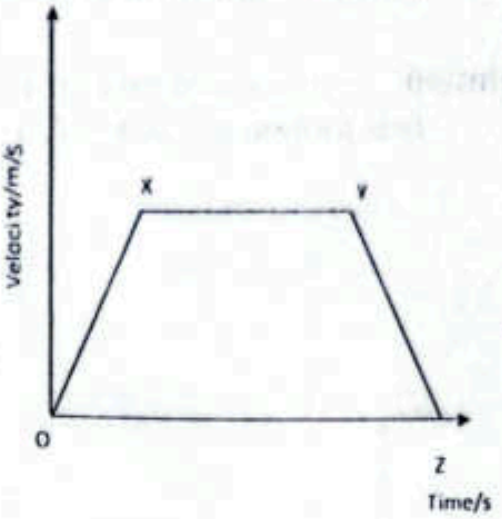


Non - uniform acceleration

Velocity time graph



Uniform velocity



Uniform acceleration

OX - represents uniform acceleration
XY - represents constant velocity or no acceleration
YZ - represent uniform deceleration
The slope of the velocity- time graph represents the *acceleration* and the area under the curve represents *the distance covered*

Solved examples

6 A particle starts from rest and accelerates to velocity of 3m/sin 10s. It maintains this velocity for 10s and then decelerates to rest in a further 10s.

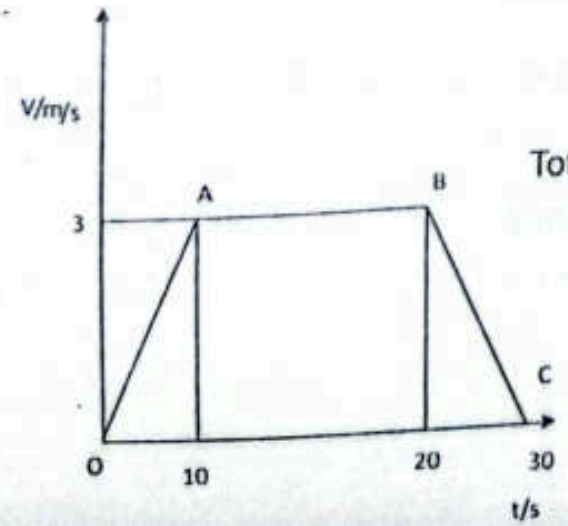
i .Draw a velocity-time graph of the motion.

ii. Calculate the initial acceleration of the particle and the total distance covered.

Solution

$$\text{Initial acceleration} = \frac{\Delta V}{\Delta t} \quad \frac{AE}{OE}$$

$$= \frac{3-0}{10-0} = 0.3\text{m/s}^2$$



Total distance covered = area under the graph

$$\text{Total distance covered} = \frac{1}{2}(oc + AB) \times 3$$

$$= \frac{1}{2}(30 + 10) \times 3 = 60\text{m}$$

7. A body starts from rest and accelerates to 3 m/s² for 5s. It maintains this velocity for 10s and then decelerates to rest in 5s.

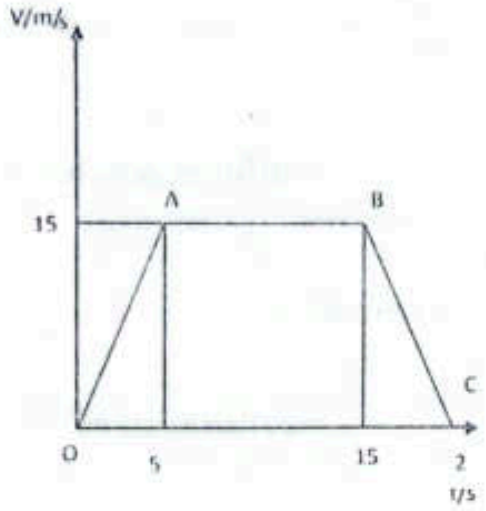
i. Sketch velocity time graph for the motion

ii. Calculate the distance covers the body

Solution

i. first motion $u = 0$, $v = ?$, $a = 3m/s^2$, $t = 5s$

$$V = 3 \times 5 = 15m/s$$



Distance covered by body

$$s = \frac{1}{2}(a + b)h$$

$$= \frac{1}{2}(OC + AB) \times 15$$

$$\frac{1}{2}(20 + 10) \times 15 = 225m$$

3.2.2 Instantaneous motion

This is the study of an object in motion at a specific point in time. *The instantaneous velocity is defined as the limit of average velocity as the time interval tends to zero.* It is also described as the velocity of an object under motion at a specific point of time

$$V_{inst} = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx(t)}{dt} \quad \ldots\ldots\ldots\ldots\ldots i$$

The instantaneous acceleration a_{inst} *is defined as the limit of average acceleration* a_{av} *as the time interval* Δt *tends to zero*

$$a_{inst} = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv(t)}{dt} = \frac{d^2 x(t)}{Dt} \quad \ldots\ldots\ldots\ldots ii$$

From equation i and ii

$$X(t) = \int_t^t v(t)dt + cand$$

$$v(t) = \int_t^t a(t)dt + c$$

Where c is a constant.

t/s

Solved examples

8. The acceleration with which a particle moves is given as $a(t) = (-5+3t^2)\ m/s^2$. Find its velocity at time t = 4s, given that the velocity at time t = 0 is zero.

Solution

The velocity of the moving particle can be obtained by integrating $a(t) = (-5+3t^2)\ m/s^2$

$$v(t) = \int_{t_1}^{t_2} a(t)dt + c$$

$$V(t) = \int (-5+3t^2)dt = -5t+t^3+c \quad \ldots\ldots\ldots\ldots\ldots\ldots (i)$$

To obtain the constant c, recall that the initial velocity was zero at time t = 0. Then

$$0 = -5 \times 0 + 0^3 + c$$

$$C = 0$$

The velocity at Hence, t = 4s can be obtained by substituting c and t = 4s into(i)

$$v = (-5 \times 4) + 4^3 + 0$$

$$v = 44 m/s$$

9. An elementary particle is projected into space and travelled as

$$r = (t^4 + 10t)\,\mathbf{i} - 5e^{-3t}\mathbf{j} + 3\sin 4t\mathbf{k}$$

Find its acceleration at t = 0

Solution

$$r = (t^4 + 10t)\,\mathbf{i} - 5e^{-3t}\mathbf{j} + 3\sin 4t\mathbf{k}$$

Differentiation of displacement with respect to t gives Velocity, then

$$v = \frac{dr(t)}{dt}$$

$$v = (t^3 + 10)\,\mathbf{i} - 15e^{-3t}\mathbf{j} + 12\cos 4t\mathbf{k}$$

Differentiation of Velocity with respect to t gives Acceleration, then

$$a = \frac{dv(t)}{dt} = \frac{d^2r(t)}{dt^2}$$

$$= (12t^2)\mathbf{i} - 45e^{-3t}\mathbf{j} - 48\sin 4t\mathbf{k}$$

At t = 0

$$a = (12(0^2)\mathbf{i} - 45e^{-3t(0)}\mathbf{j} - 48\sin 4(0)\,\mathbf{k}$$

$$a = (-45j) m/s^2$$